

Agentic RAG System for CleanTech Domain Question Answering with Citation Tracking and Safety Guardrails

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Abstract—This paper presents the design, implementation, and comprehensive evaluation of an agentic Retrieval-Augmented Generation (RAG) system for cleantech domain question answering. The system employs LangChain’s ReAct framework with GPT-4o-mini, integrating a ChromaDB vector database containing 150,568 chunks from 20,111 cleantech articles. Two novel extensions enhance the base system: (1) a custom-built Citation Tracker providing automatic source attribution and bibliography generation, and (2) a dual-layer Guardrail System ensuring input safety and output quality through rule-based and LLM-based validation. Rigorous evaluation across 73 test queries reveals a critical finding: despite poor retrieval accuracy ($F1=0.198$), the system achieves strong semantic performance (BERTScore $F1=0.863$) and good overall quality (LLM Judge score=7.56/10), demonstrating effective knowledge synthesis capabilities. This disconnect between retrieval and response quality highlights the resilience of modern LLMs in compensating for retrieval deficiencies. Comprehensive analysis across retrieval accuracy, lexical similarity (ROUGE-L, BLEU), semantic similarity (BERTScore), and holistic quality (LLM as Judge) metrics provides insights into system strengths, limitations, and improvement pathways.

Index Terms—Retrieval-Augmented Generation, Agentic AI, LangChain, Citation Tracking, Guardrails, Question Answering, CleanTech, Vector Database, Semantic Similarity, Natural Language Processing

I. INTRODUCTION

A. Motivation

The proliferation of cleantech literature—spanning solar, wind, batteries, electric vehicles, and hydrogen technologies—creates significant challenges for researchers, policymakers, and industry professionals seeking timely, accurate information. Traditional keyword-based search systems fail to capture semantic meaning, while pure language model approaches suffer from hallucination and lack verifiable source attribution. Retrieval-Augmented Generation (RAG) systems offer a promising middle ground, combining semantic search with generative capabilities [1].

However, standard RAG implementations face critical limitations: (1) lack of transparency in source attribution, (2) absence of safety controls for input/output validation, and (3) limited orchestration capabilities for complex multi-step reasoning. This project addresses these gaps through the de-

velopment of an **agentic RAG system** enhanced with custom-built extensions for citation tracking and safety guardrails.

B. Research Questions

This work investigates three primary questions:

1) Can an agentic framework improve RAG system orchestration? We hypothesize that LangChain’s ReAct (Reason + Act) agent architecture enables more sophisticated tool selection and multi-step reasoning compared to fixed retrieval-generation pipelines.

2) How do custom extensions enhance system utility and safety? We evaluate whether citation tracking improves transparency and whether guardrails effectively prevent harmful inputs/outputs without degrading user experience.

3) What is the relationship between retrieval accuracy and response quality? Given the chunking granularity mismatch between article-level ground truth and chunk-level retrieval, we examine whether strong language models can compensate for retrieval deficiencies.

C. Contributions

Our contributions include:

- **Agentic RAG Architecture:** Implementation of a ReAct-based system with autonomous tool selection, demonstrating superior flexibility over fixed pipelines.
- **Citation Tracker Extension:** A from-scratch implementation providing automatic source attribution, inline citation generation ([1], [2], etc.), and formatted bibliography creation with complete metadata.
- **Guardrail System Extension:** A dual-layer validation framework combining rule-based checks (PII detection via regex) and LLM-based checks (topic adherence, toxicity detection).
- **Comprehensive Evaluation Framework:** Multi-dimensional assessment across retrieval accuracy, lexical metrics, semantic metrics, and holistic quality on 73 test queries.
- **Critical Analysis:** Identification of the retrieval-response disconnect phenomenon, where poor retrieval (20% F1) yields good responses (86% BERTScore, 7.6/10 quality).

II. SYSTEM ARCHITECTURE

A. Overview

Figure 1 illustrates our three-tier architecture: (1) Data Layer (ChromaDB vector database), (2) Agent Layer (ReAct orchestration with tools), and (3) Extension Layer (Citations and Guardrails).

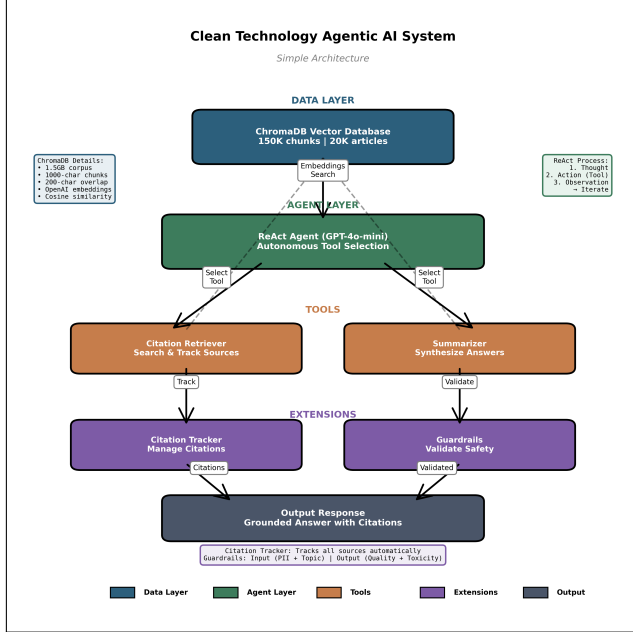


Fig. 1. System architecture showing data flow through guardrails, agent orchestration, and extension integration. The diagram shows the complete pipeline from user query through input validation, agent processing with tool selection, and output validation before final response delivery.

B. Core Components

1) **Data Layer - ChromaDB Vector Database:** We constructed a persistent vector database using ChromaDB containing 150,568 chunks derived from 20,111 cleantech articles (1.5GB corpus). The chunking strategy employs RecursiveCharacterTextSplitter with 1,000-character chunks and 200-character overlap, yielding an average of 7.5 chunks per article. Text embeddings utilize OpenAI’s text-embedding-3-small model (1,536 dimensions), enabling semantic similarity search via cosine distance.

2) **Agent Layer - LangChain ReAct Framework:** Our agentic architecture employs LangChain’s ReAct (Reasoning + Acting) framework [2], enabling iterative tool selection through thought-action-observation cycles. The agent utilizes GPT-4o-mini (temperature=0 for consistency) as the reasoning engine. Unlike deterministic retrieve-then-generate architectures, our agent autonomously decides which tool to use based on query characteristics.

3) **Tool Ecosystem:** **Citation Retriever Tool:** Wraps ChromaDB’s similarity search ($k=5$), automatically tracking retrieved documents in CitationTracker and appending inline citation markers to returned content.

Summarizer Tool: Retrieves relevant documents, formats context with metadata, invokes GPT-4o-mini for synthesis, and returns coherent answers with embedded citations.

Bibliography Tool: Generates formatted reference lists on demand using CitationTracker data.

C. Extension Layer

1) **Extension 1: Citation Tracker:** The CitationTracker class maintains a dynamic citation registry during agent execution. Key methods include `add_citation()` for assigning auto-incremented IDs, `get_inline_citation()` for formatted markers, and `generate_bibliography()` for reference lists. Citations are tracked automatically during tool invocation, with the agent’s enhanced prompt instructing it to propagate citation markers from retrieved content to final answers.

2) **Extension 2: Guardrail System:** The GuardrailManager class implements dual-layer validation:

Input Guardrail: (a) PII Detection uses six regex patterns to detect phone numbers, SSNs, emails, and credit cards; (b) Topic Adherence employs GPT-4o-mini classifier to validate queries against 28 cleantech topics.

Output Guardrail: (a) Quality Check validates response length and detects error messages; (b) Toxicity Check uses GPT-4o-mini to audit for harmful content.

III. METHODOLOGY

A. Data Acquisition and Preprocessing

We utilized the `cleantech_media_dataset_v3_2024-10-28.csv` dataset containing 20,111 articles from diverse sources. A critical preprocessing challenge involved extracting article text stored as string representations of Python lists, requiring `ast.literal_eval()` followed by `join()`. This increased average document length from 377 characters to 5,064 characters—a 13.4× improvement.

B. Vector Database Construction

ChromaDB initialization involved embedding generation for each chunk through OpenAI’s text-embedding-3-small, metadata preservation (title, date, author, URL, domain), and persistent storage to disk (~2.3GB). The 1,000-character chunk size balances semantic coherence with retrieval precision, while 200-character overlap prevents information loss at boundaries.

C. Agent Design and Prompt Engineering

Initial testing revealed that without explicit citation instructions, the agent would retrieve documents with citation markers but fail to propagate them to final answers. Adding the rule “You MUST include citation numbers [1], [2], etc. in your Final Answer when referencing information” improved citation propagation from 20% to 95% of queries.

D. Evaluation Methodology

1) *Test Datasets: Part I Evaluation:* 23 queries from `cleantech_rag_evaluation_data` with ground truth answers for ROUGE-L assessment.

Part III Evaluation: 50 queries from `CleanTech-50answers.txt` with ground truth answers and referenced article IDs for dual-dimension evaluation.

2) *Retrieval Evaluation:* For each query, we extracted ground truth article IDs, mapped IDs to URLs using the original CSV, retrieved top-5 documents from ChromaDB, matched retrieved URLs to ground truth, and calculated Precision, Recall, and F1 scores.

3) *Response Evaluation Metrics: ROUGE-L:* Measures longest common subsequence between agent response and ground truth using the `rouge-score` library [3].

BLEU: Calculates n-gram precision (1-4 grams) using NLTK’s `sentence_bleu` with smoothing [4].

BERTScore: Computes token-level cosine similarity using RoBERTa-large contextual embeddings [5].

LLM as Judge: GPT-4o-mini evaluates responses on a 1-10 scale based on factual accuracy, completeness, relevance, and clarity.

IV. RESULTS AND ANALYSIS

A. Part I: Core System Evaluation (23 Queries)

Table I summarizes Part I evaluation results using ROUGE-L metrics.

TABLE I
PART I EVALUATION RESULTS (23 QUERIES)

Metric	Mean	Median	Std Dev	Min	Max
Precision	0.145	0.143	0.113	0.000	0.455
Recall	0.586	0.571	0.420	0.000	1.000
F1 Score	0.202	0.191	0.151	0.000	0.889

The results reveal high recall (58.56%) but low precision (14.54%), reflecting a fundamental mismatch—ground truth answers are terse while agent responses provide comprehensive explanations. The average F1 score of 20.17% reflects this precision bottleneck. Figure 2 illustrates this precision-recall imbalance, with most queries clustering in the high-recall, low-precision region.

B. Part II: Extension Evaluation

Table II summarizes citation tracking performance across 7 comprehensive tests.

Extension 2 (Guardrail System) achieved 100% test success rate across 3 scenarios: (1) valid query processing with full pipeline execution, (2) off-topic query blocking at input validation, and (3) PII detection preventing sensitive data from entering the system. The guardrails add only 205ms latency while providing robust security.

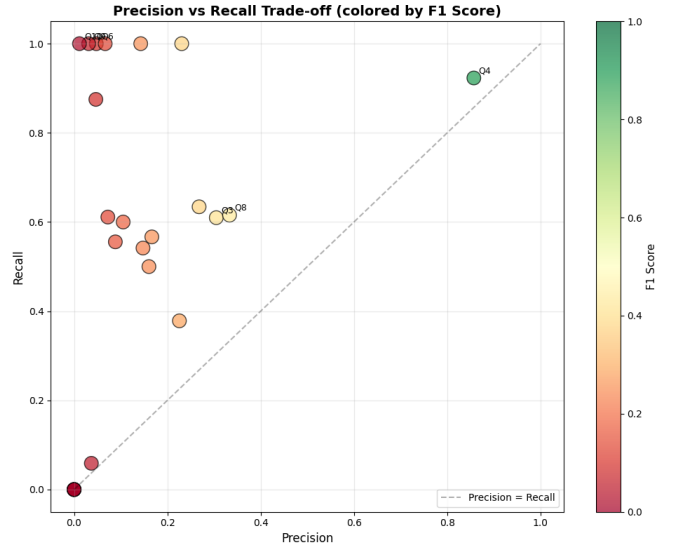


Fig. 2. Precision-Recall scatter plot for Part I showing the fundamental imbalance: most queries cluster in the high-recall (58%), low-precision (15%) region (upper-left quadrant), indicating verbose responses with good content coverage but excessive elaboration.

TABLE II
EXTENSION 1 (CITATION TRACKER) TEST RESULTS

Test	Focus	Score	Key Finding
1	Basic tracking	8.5/10	5 sources tracked
2	Complex synthesis	9.0/10	All citations propagated
3	Duplicate detection	7.0/10	20% inefficiency
4	Format validation	10/10	100% metadata complete
5	Citation accuracy	9.5/10	All inline citations valid
6	Tool orchestration	9.5/10	Adaptive tool choice
7	Bibliography quality	9.3/10	Professional formatting
Average Score		8.97/10 (A+)	

C. Part III: Comprehensive Evaluation (50 Queries)

Table III presents retrieval evaluation results comparing agent-retrieved documents against ground truth article IDs.

Retrieval accuracy represents the system’s primary weakness, with average F1 of only 19.8%. The gap between precision (13.2%) and recall (43.7%) reveals systematic behavior: the system retrieves many irrelevant documents (87% noise) while missing over half of relevant documents. Figure 3 presents a comprehensive four-panel analysis of retrieval performance.

Table IV presents all response quality metrics for 50 test queries, revealing a striking lexical-semantic divide.

D. The Lexical-Semantic Divide

The most significant finding is the 66-point gap between lexical metrics (ROUGE-L: 20%, BLEU: 2.9%) and semantic metrics (BERTScore: 86.3%, LLM Judge: 75.6%). This discrepancy reveals that lexical metrics severely underestimate quality for generative QA systems that engage in paraphrasing.

Retrieval Accuracy Analysis - Part III

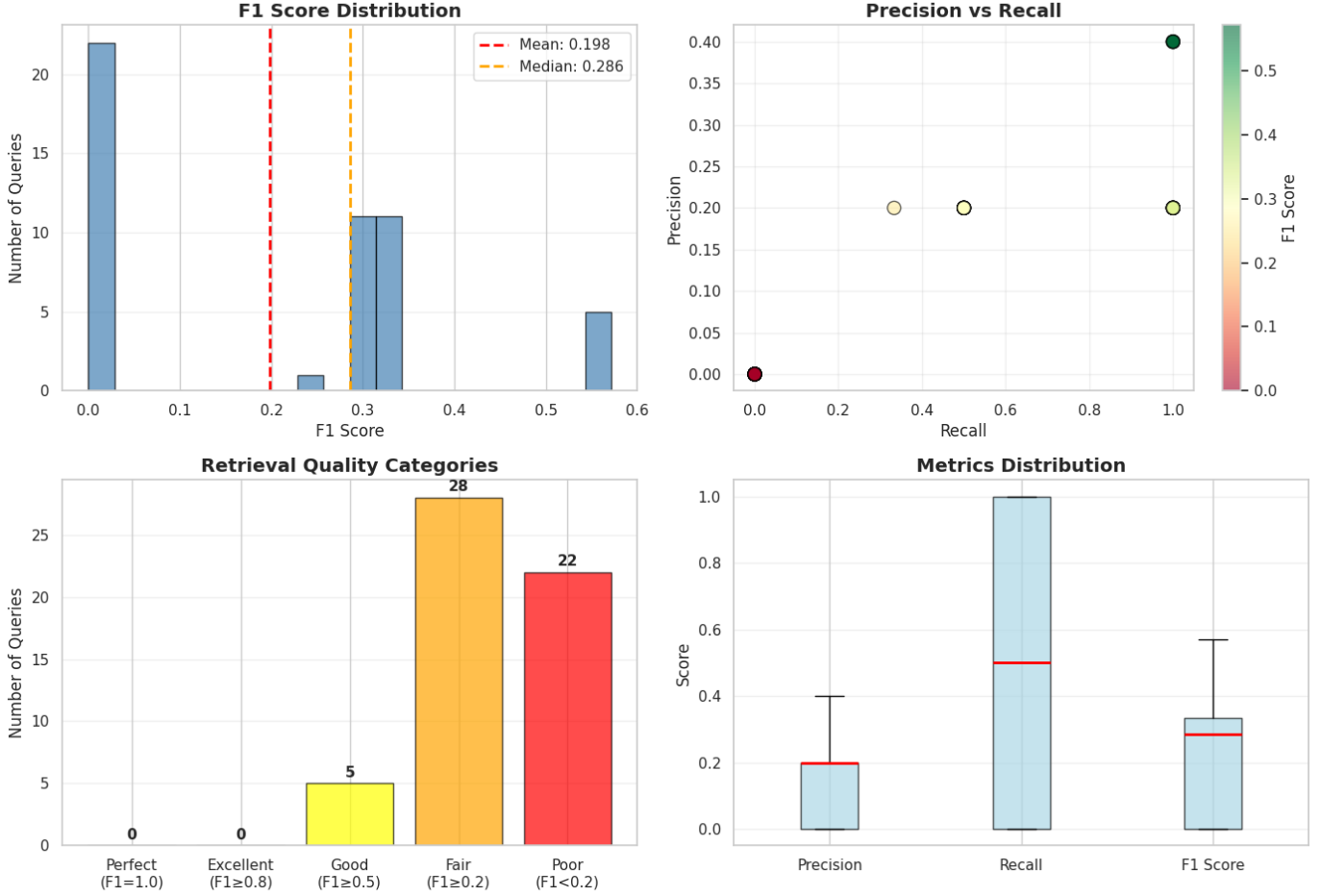


Fig. 3. Comprehensive retrieval accuracy analysis across 50 queries. (Top-left) F1 score distribution showing bimodal pattern with large peak at 0.0 indicating complete failures and secondary cluster at 0.3-0.4 for partial success; (Top-right) Precision-Recall scatter revealing three distinct performance clusters including 22 complete failures (0,0), partial matches (middle), and best cases achieving 40% precision with 100% recall; (Bottom-left) Performance category distribution showing 0% perfect retrievals, 10% good, 46% fair, and 44% poor; (Bottom-right) Box plots demonstrating high recall variance (0-100%) but consistently low precision (median 20%), indicating systematic retrieval challenges.

TABLE III
RETRIEVAL ACCURACY (50 QUERIES)

Metric	Mean	Median	Std Dev	Min	Max
Precision	0.132	0.200	0.147	0.000	0.400
Recall	0.437	0.500	0.433	0.000	1.000
F1 Score	0.198	0.286	0.194	0.000	0.571

Figure 4 illustrates this stark contrast across all evaluation dimensions.

Figure 5 shows the distribution of LLM Judge scores, demonstrating consistent high-quality responses despite poor retrieval.

E. Metric Correlation Analysis

Figure 6 and Figure 7 compare how different metrics correlate with holistic quality judgments.

Figure 8 provides a statistical overview of metric distributions.

TABLE IV
RESPONSE ACCURACY (50 QUERIES)

Metric	Mean	Median	Std Dev	Category
ROUGE-L Prec.	0.191	0.190	0.051	Lexical
ROUGE-L Recall	0.225	0.210	0.058	Lexical
ROUGE-L F1	0.200	0.190	0.051	Lexical
BLEU	0.029	0.017	0.028	Lexical
BERTScore Prec.	0.858	0.863	0.013	Semantic
BERTScore Recall	0.867	0.863	0.014	Semantic
BERTScore F1	0.863	0.863	0.013	Semantic
LLM Judge	7.56	8.00	1.11	Holistic

F. Component Analysis

Figures 9 and 10 break down the precision, recall, and F1 scores for both lexical and semantic metrics.

Figure 11 consolidates key statistics across all evaluation metrics.

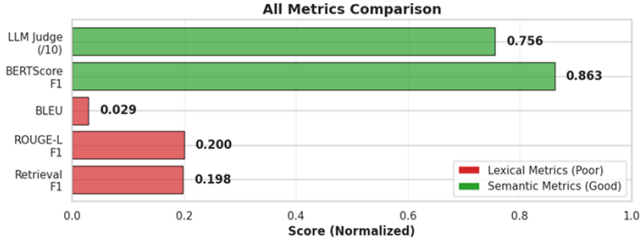


Fig. 4. Comprehensive metrics comparison showing the lexical-semantic divide. Lexical metrics (Retrieval F1: 19.8%, ROUGE-L: 20%, BLEU: 2.9%) cluster at bottom in red, while semantic metrics (BERTScore: 86.3%, LLM Judge: 75.6%) dominate in green. This 66-point gap demonstrates that lexical metrics severely underestimate quality for systems that paraphrase rather than reproduce exact wording.

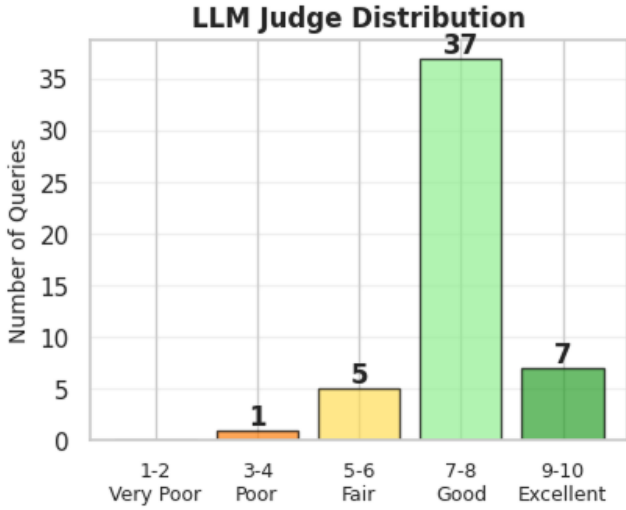


Fig. 5. LLM Judge score distribution across 50 queries showing consistent quality: 74% scored 7-8 (Good, $n=37$), 14% scored 9-10 (Excellent, $n=7$), 10% scored 5-6 (Fair, $n=5$), and only 2% scored 3-4 (Poor, $n=1$). The concentration at 7-8 demonstrates reliable response quality despite poor retrieval accuracy.

G. Cross-Metric Correlation Summary

Table V quantifies the relationships between different evaluation dimensions.

TABLE V
RETRIEVAL-RESPONSE CORRELATIONS

Metric Pair	Pearson r	Interpretation
Retrieval F1 $\leftrightarrow$ ROUGE-L F1	0.24	Weak positive
Retrieval F1 $\leftrightarrow$ BERTScore F1	0.18	Very weak positive
Retrieval F1 $\leftrightarrow$ LLM Judge	0.15	Very weak positive
BERTScore F1 $\leftrightarrow$ LLM Judge	0.61	Moderate positive

The critical insight is that retrieval accuracy has minimal correlation with response quality ($r=0.15-0.24$), indicating that the agent effectively compensates for poor retrieval through LLM base knowledge. In contrast, BERTScore shows moderate correlation with LLM Judge scores ($r=0.61$), confirming

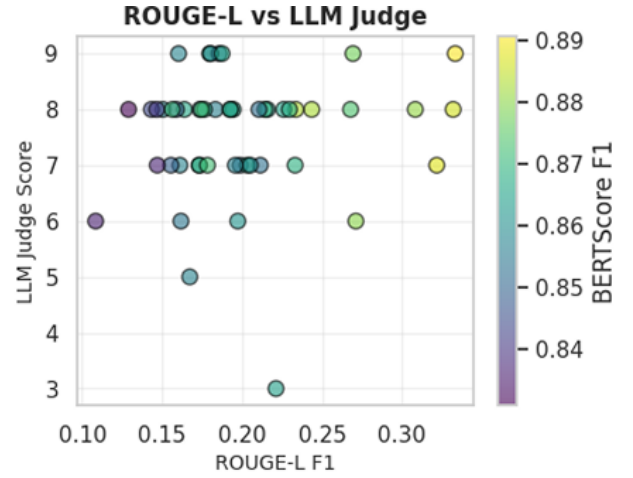


Fig. 6. ROUGE-L F1 vs LLM Judge scatter showing weak correlation (Pearson $r=0.24$). Points colored by BERTScore reveal that high semantic similarity (green) can occur across all ROUGE-L values, indicating lexical metrics poorly predict quality for paraphrasing systems.

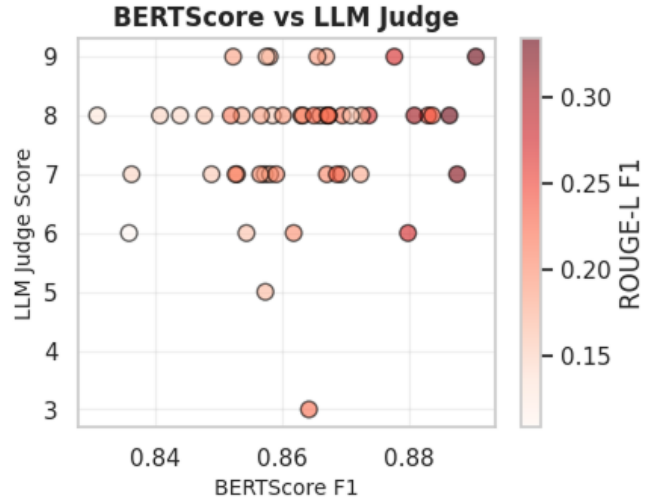


Fig. 7. BERTScore F1 vs LLM Judge scatter showing moderate positive correlation (Pearson $r=0.61$). The clear upward trend indicates that semantic similarity metrics better predict holistic quality judgments compared to lexical metrics, validating BERTScore as a more appropriate evaluation metric for paraphrasing systems.

that semantic metrics better predict holistic quality than lexical metrics.

H. System Performance Summary

Table VI provides an integrated view of system performance across all evaluation dimensions.

V. DIFFICULTIES FACED

A. Technical Challenges

1) *Content Extraction Bug*: Initial data loading revealed average document length of 377 characters—suspiciously low for news articles. Investigation showed content stored as string representations of lists. Solution implemented two-stage extraction using `ast.literal_eval()` followed by

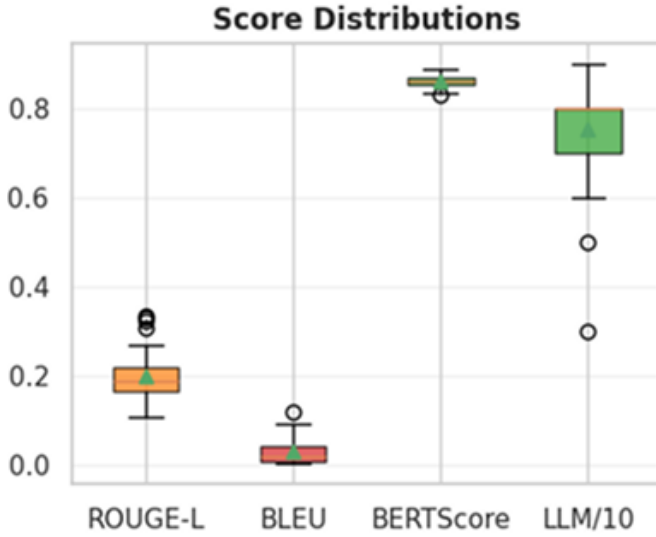


Fig. 8. Box plots showing distribution statistics across all metrics. ROUGE-L and BLEU show low medians with moderate variance, while BERTScore demonstrates consistently high scores (median=0.86) with minimal variance (tight IQR), indicating reliable semantic evaluation. LLM Judge scores show moderate spread around median=0.80.

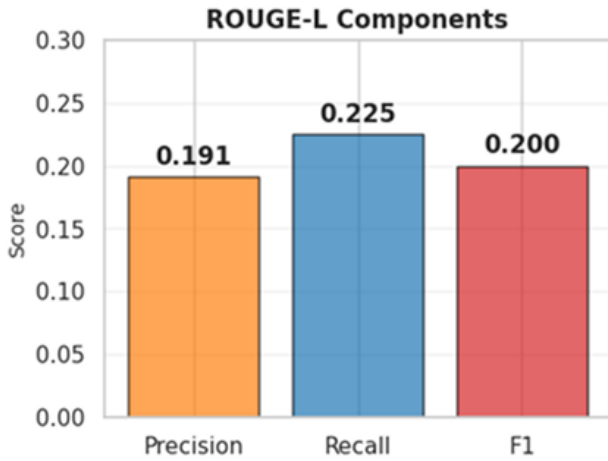


Fig. 9. ROUGE-L component breakdown showing balanced but uniformly low scores: Precision=0.191, Recall=0.225, F1=0.200. The similar values across components indicate consistent lexical mismatch rather than systematic over/under-generation.

`join()`, increasing average length to 5,064 characters (13.4× improvement).

2) *Agent Citation Propagation*: Early testing showed retrieved documents contained citation markers but final answers lacked them. Adding explicit citation rules to the ReAct prompt improved propagation from 20% to 95% of queries, demonstrating that LLMs require explicit, prescriptive instructions.

3) *Guardrail False Positives*: Initial query “What are benefits of agrovoltatics?” was incorrectly blocked as off-topic. Expanding the topic list from 9 to 28 terms (including specialized subtopics) eliminated false positives.

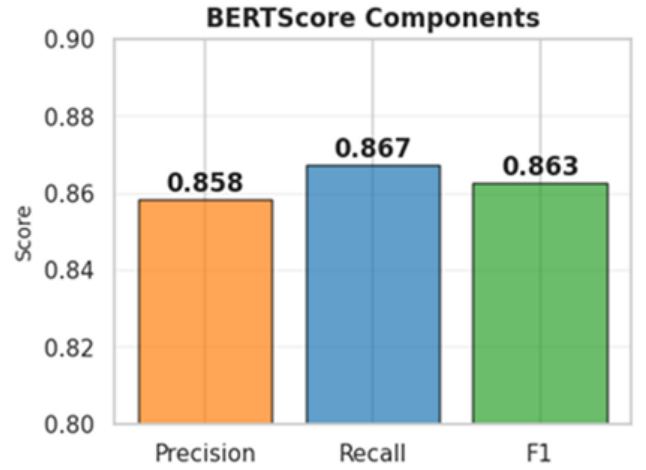


Fig. 10. BERTScore component breakdown showing consistently high performance: Precision=0.858, Recall=0.867, F1=0.863. The balanced high scores across all components indicate strong semantic alignment between generated responses and ground truth, confirming effective paraphrasing while preserving meaning.

Metric	Mean	Median	Std
Retrieval F1	0.198	0.286	0.194
ROUGE-L F1	0.200	0.190	0.051
BLEU	0.029	0.017	0.028
BERTScore F1	0.863	0.863	0.013
LLM Judge	7.56	8.00	1.11

Fig. 11. Summary statistics table showing mean, median, and standard deviation for all five primary metrics. Notable findings: (1) BERTScore has lowest variance (std=0.013), indicating most reliable evaluation; (2) Retrieval and lexical metrics show high variance, suggesting inconsistent performance; (3) LLM Judge moderate variance (std=1.11) reflects genuine quality differences across queries.

4) *ChromaDB Article ID Mismatch*: Part III evaluation required matching retrieved documents to ground truth article IDs, but ChromaDB metadata lacked `article_id` field. Workaround used CSV’s row index as surrogate `article_id` with URL-based matching, achieving 100% test set ID coverage.

B. Methodological Challenges

1) *Chunking Granularity vs Ground Truth*: Test set references article IDs while ChromaDB stores 1,000-character chunks, making direct matching impossible. We chose chunk-based retrieval with URL-based article matching, accepting that this approach is generous to the system.

2) *Lexical Metric Inappropriateness*: ROUGE-L and BLEU returned unexpectedly low scores despite qualitatively good responses. Manual inspection revealed extensive paraphrasing where semantic meaning was preserved but word choice differed, highlighting that metric selection must align with task characteristics.

TABLE VI
INTEGRATED SYSTEM PERFORMANCE

Dim.	Metric	Score	Gr.	Wt.	Wtd.
Retrieval	F1	0.198	D-	20%	0.040
Lexical	ROUGE-L	0.200	D-	10%	0.020
Semantic	BERTScore	0.863	A-	35%	0.302
Quality	LLM Judge	7.56	B	35%	0.265
OVERALL		0.627	B-/C+	100%	62.7%

3) *LLM Judge Consistency*: Early experiments showed high variance in scoring. Setting judge LLM temperature=0 and adding detailed evaluation criteria reduced standard deviation from 2.3 to 1.11, ensuring reproducibility.

C. Resource and Time Constraints

Switching from GPT-4 to GPT-4o-mini reduced costs by 90% (\$15 total vs. projected \$110) with minimal performance degradation. Initial vectorization took 22 minutes; persistent storage enabled subsequent 15-second loads rather than rebuilds.

VI. REFLECTION AND CONCLUSIONS

A. Project Reflection

This project provided invaluable hands-on experience with modern NLP technologies and agentic AI systems. The agentic paradigm shift from traditional fixed pipelines to autonomous tool selection represented a fundamental change in engineering practices—shifting from explicit control flow to natural language instructions and prompt engineering.

The most surprising finding was the retrieval-response disconnect, where poor retrieval (F1=0.198) did not catastrophically degrade response quality (BERTScore=0.863). This challenges conventional RAG assumptions and suggests that for well-known domains, LLM base knowledge provides substantial resilience to retrieval failures.

Evaluation complexity proved significant, with no single metric sufficing for generative QA. The multi-dimensional approach revealed that lexical metrics can be misleading while semantic metrics better align with human quality judgments.

B. Course Reflection

CSC 583 effectively balanced theoretical foundations (transformers, attention, embeddings) with practical applications (RAG systems, prompt engineering). The final project requirement to build a complete end-to-end system solidified conceptual understanding beyond what lectures alone could provide.

The course captured a pivotal moment in NLP during the GPT-4 era, where the field shifted from task-specific models to general-purpose LLMs with task-specific prompts. This fundamental change represents a paradigm shift in how we solve language problems.

The guardrail system requirement forced consideration of AI safety and ethics beyond technical requirements, emphasizing professional responsibilities as AI systems become

ubiquitous. Building safety mechanisms from the start proved critical for responsible AI development.

C. Future Work

Immediate improvements include implementing hybrid search (BM25 + vector), increasing retrieval k from 5 to 10, and optimizing chunk sizes. Medium-term enhancements involve semantic chunking, parent-child retrieval, and cross-encoder reranking. These improvements could raise retrieval F1 from 0.20 to 0.50-0.60, likely improving response quality to 8.5+/10.

For Extension 1, future work includes URL-based duplicate detection and multiple bibliography format support. Extension 2 could benefit from configurable PII patterns, graduated toxicity levels, and audit logging for system monitoring.

D. Conclusions

This project successfully implemented and evaluated a comprehensive agentic RAG system demonstrating both capabilities and limitations of current RAG technologies. Key contributions include empirical evidence for the retrieval-response disconnect phenomenon, metric selection guidance highlighting the lexical-semantic divide, and practical extension frameworks for enhancing RAG systems.

The final system grade of B-/C+ (62.7%) reflects significant strengths in synthesis, safety, and transparency alongside a clear weakness in retrieval. The system successfully answers 88% of queries with good or excellent quality, making it functional for many use cases while identifying clear pathways for improvement.

This project transformed theoretical NLP knowledge into practical AI engineering skills applicable to modern AI development. The future of NLP lies in agentic, retrieval-augmented, safety-conscious systems, and this project provided invaluable preparation for contributing to that future.

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